

# **Path Dependence and Growth: The Role of Oil in Developing Economies**

Toribio Iriarte

i6317519

EBC2090

Advisor: Julian Ashwin

21<sup>st</sup> of January 2025

Word Count: 10201

Number of Pages: 37



**Maastricht  
University**

## Abstract

This paper examines the path-dependent dynamics of oil in developing countries, focusing on the causal relationship between institutional quality and economic output. Using a two-stage least squares (2SLS) regression, the study employs a 10-year lag of oil resources as an instrumental variable to address endogeneity concerns, isolating the effect of oil on institutions and GDP per capita. Institutions are captured using a principal component analysis of five key measures: corruption, absence of political violence, transparent laws, bureaucratic quality, and property rights. Economic growth dynamics are modeled through an augmented Solow framework, accounting for the role of nonrenewable resources and fixed land.

The results validate the instrument's relevance, with a high first-stage F-statistic confirming its strength and robustness. The first-stage findings show that oil negatively impacts institutional quality, consistent with resource curse theory. In the second stage, the instrumented institutions also exert a significant negative effect on GDP per capita, suggesting a detrimental path-dependent relationship between institutions and oil, negatively impacting per capita output. Robustness checks incorporating additional controls, namely unemployment, precipitation, and temperature, further ensure the validity of the instrument and, as a result, the main findings.

In the second phase of analysis, colony interaction terms provide insights into the interaction of oil path dependence and colonial legacies. French and Portuguese colonial histories are shown to exacerbate oil's negative impact on institutions, and the negative institutional impact on economic output, while Dutch colonies demonstrate a mitigating effect. British colonies and nations without colonial histories exhibit intermediate effects, with Russian and Spanish legacies displaying less detrimental interactions than the former, but still amplifying oil's negative path-dependent impact.

In the final phase of analysis, the disaggregation of institutional components reveals the specific institutional mechanisms driving these dynamics. Corruption emerges as the dominant channel through which oil undermines institutional development and subsequently hinders economic output. Bureaucratic quality, transparent laws, and absence of political violence also exhibit negative effects but to a lesser degree, while property rights yield statistically insignificant results due to weak instrument relevance.

These results suggest that natural wealth negatively conditions institutional development, supporting the contention that institutions are path-dependent on oil endowments. By establishing a causal relationship between institutions and economic output, this paper provides empirical evidence for the detrimental long-term impacts of oil wealth on developing countries' growth trajectories.

# Contents

|          |                                                     |           |
|----------|-----------------------------------------------------|-----------|
| <b>1</b> | <b>Research Overview</b>                            | <b>4</b>  |
| <b>2</b> | <b>Literature Review</b>                            | <b>5</b>  |
| 2.1      | Institutional Theory . . . . .                      | 5         |
| 2.2      | Path Dependence . . . . .                           | 6         |
| 2.3      | Economic Growth Models . . . . .                    | 7         |
| 2.4      | Causal Identification through Instruments . . . . . | 8         |
| <b>3</b> | <b>Methodology</b>                                  | <b>11</b> |
| 3.1      | Data . . . . .                                      | 14        |
| 3.2      | Institution Aggregation . . . . .                   | 16        |
| 3.3      | Augmented Solow Model . . . . .                     | 17        |
| 3.4      | Econometric Setup . . . . .                         | 19        |
| 3.4.1    | First Stage . . . . .                               | 19        |
| 3.4.2    | Second Stage . . . . .                              | 20        |
| 3.4.3    | Robustness Check . . . . .                          | 20        |
| 3.4.4    | Colonial Impact . . . . .                           | 21        |
| 3.4.5    | Disaggregated Institutions . . . . .                | 22        |
| 3.5      | Instrumental Variable . . . . .                     | 23        |
| <b>4</b> | <b>Analysis</b>                                     | <b>25</b> |
| 4.1      | Main Results . . . . .                              | 25        |
| 4.2      | Colonial Impact . . . . .                           | 30        |
| 4.3      | Institution Disaggregation . . . . .                | 34        |
| <b>5</b> | <b>Caveats and Study Considerations</b>             | <b>38</b> |
| <b>6</b> | <b>Conclusion</b>                                   | <b>40</b> |
| <b>A</b> | <b>Appendix</b>                                     | <b>41</b> |
| <b>B</b> | <b>Bibliography</b>                                 | <b>47</b> |
| <b>C</b> | <b>Statement of Original Thesis</b>                 | <b>50</b> |

# 1 Research Overview

Natural endowment has long been considered a blessing for countries, providing comparative advantages in trade that drive development and growth. However, economic thought has evolved beyond this conventional view, contending that resource abundance is often a net detriment to nations due to its currency effects and impact on institutional quality. Known as the Dutch Disease and Resource Curse, these theories suggest that natural endowments may serve as a hindrance to growth and development, particularly in developing nations. Research shows that resource-rich nations often grow more slowly than resource-scarce ones due to the erosion of institutional quality (Sachs and Warner, 2005; Ross, 2001). These dynamics raise critical questions about the long-term impact of natural resources on institutional development, particularly through path-dependent processes that shape formal institutions.

The concept of path dependence offers a valuable perspective on how resource endowment shapes institutional trajectories. Colonial powers frequently established extractive institutions to exploit natural wealth, creating legacies that continue to shape governance structures, economic policies, and development. In resource-rich countries, these colonial legacies are thought to exacerbate the institutional vulnerabilities linked to resource dependence, making them a critical factor to consider. While resource endowments affect countries differently, oil stands out as a commodity with uniquely significant economic and institutional implications. Some nations, such as Saudi Arabia and the UAE, have successfully leveraged oil wealth for rapid growth and infrastructure development, while others have experienced economic crises linked to volatile commodity prices. These differences underscore the need to examine resource-specific impacts and the conditions under which resource wealth either fosters or impedes development.

This paper investigates the presence of institutional path dependence in oil-rich countries. Specifically, it poses the question: Are institutions in developing countries path dependent on oil, and does this impact the causal relationship between institutions and economic output? To address this and establish causality, an instrumental variable approach is employed. Through a two-stage least squares framework, this study investigates the relationship between oil wealth and institutional quality, as well as the causal impact of instrumented institutions on economic output. The analysis is further extended to examine the interaction between oil path dependence and colonial legacies, exploring whether colonial histories amplify or mitigate the path-dependent effects of oil wealth. Lastly, a disaggregation of institutional measures will be carried out to identify the specific institutional dimensions most affected by oil.

The findings confirm a negative path-dependent relationship between oil and institutional quality, supporting the negative causal impact of oil-instrumented institutions on GDP per capita. Colonial interactions are significant, with both exacerbating and moderating effects on oil path dependence. Corruption emerges as the dominant mechanism driving these dynamics, highlighting the challenges oil wealth poses to both institutions and economic growth.

## 2 Literature Review

A substantial portion of economic literature has been dedicated to studying growth and development, perceived by many as the foremost concern of economics. In this effort, various concepts such as path dependence and institutional development have been proposed as primary drivers and determinants of growth. This paper seeks to build upon these foundational concepts to explore the interplay between oil, institutions, and economic growth. Accordingly, this literature review is organized around four core themes: institutional theory, path dependence, causal identification through instruments, and economic growth models. Through a systematic review of the existing scholarship, this section establishes the conceptual foundation and empirical insights that underpin the analysis of this study.

### 2.1 Institutional Theory

The foundational role of institutions in shaping economic outcomes has been of the most influential bodies of literature in economics. Defined by North (2005) as the "rules of the game" structuring interactions between individuals and collective actors, these institutions have been identified as potential accelerators or hindrances to economic growth, through channels such as property rights, bureaucracy and democracy.

In Acemoglu and Johnson's (2005) seminal work "Unbundling Institutions", the authors seek to distinguish between property rights institutions and contracting institutions. Using colonial history as an instrumental variable to isolate the causal effect of these institutional types, the authors find that property rights institutions have a stronger and more direct effect on economic growth, investment, and financial development. Contracting institutions, on the other hand, were found to have weaker impacts on macroeconomic outcomes, underscoring the primacy of property rights in fostering economic development. This study highlights the critical role of political and institutional structure in economic growth, providing a framework for instrumental variable analysis in institutional development. It also establishes a causal relationship between institutions and economic growth, setting the empirical foundation for this study.

Frericks et al. (2018) lay out critiques of existing methods of institutional analysis in their paper "The Difficulty of Measuring Institutions". The authors propose a method of comparative measurement of both qualitative and quantitative data regarding institutions, allowing a systematic evaluation of institutional characteristics using benchmarks. This study emphasizes the benefits of using a continuous scale in the measurement of institutions, allowing for nuanced differentiation rather than binary distinctions. The use of observable and measurable institutions was also stressed, as measuring property rights and bureaucracy is more practical than informal norms that permeate society. Furthermore, it highlights the value of the aggregation of institutional measures in empirical research, facilitating meaningful comparisons of institu-

tional development across countries and over time. By bridging the gap between narrative and quantitative analysis, this study presents a robust framework that supports the use of aggregated institutional indicators to enhance cross-country and temporal assessments.

Leftwich (1993) explores the challenges of measuring institutions in developing countries in the paper titled "Governance, Democracy and Development in the Third World". This study critiques the assumption that governance and democracy are essential preconditions for development, highlighting that managerial efficiency is often conflated with democratic systems. A comparative study of the historical role of centralized states in successful developmental transformations reveals that development frequently occurs under non-democratic or semi-democratic regimes, challenging the notion that democracy is a prerequisite for growth. The author underscores the importance of "developmental states," which prioritize strategic economic goals and implement reforms to drive growth. This analysis is particularly relevant when developing composite measures for institutional quality in developing economies, as governance structures vary significantly, shaped by historical, cultural, and political contexts. This diversity poses challenges in accurately measuring institutional effectiveness, particularly in identifying the specific factors that drive growth and development.

## **2.2 Path Dependence**

Path dependence highlights the persistent influence of historical events and decisions on the evolution of societal processes and outcomes (Ernst & Preston, 2017). In the context of economics, path dependence has been used to explain how historical events, such as colonial history, have shaped the evolution of institutions. This concept can also be leveraged to understand how resource wealth, particularly oil, influences institutional trajectories and economic growth.

A study titled "Oil and Property Rights" by De Soysa et al. (2022) examines the institutional resource curse, providing insights into how oil production undermines the protection of property rights. Using instrumental variables, the study demonstrates that oil wealth fosters concentrated interest groups with disproportionate economic and political power, finding that higher oil income per capita correlates with weaker property rights. The study demonstrates the utility of instrumental variable methods in resource-based institutional analyses, making a significant contribution to the understanding of how oil wealth shapes institutional quality and impacts economic outcomes in the field of economics.

Paul A. David (2007) lays a foundational understanding of path dependence in the context of economics in the paper "Path Dependence: Conceptual Foundations." Through an analysis of non-ergodic systems, where initial conditions and random perturbations exert lasting influence, the paper provides insights into the role of path dependence in institutional evolution and economic growth. The key mechanisms of path dependence identified by the author include positive feedback loops, lock-in effects, and critical junctures. Positive feedback loops include

initial events such as the establishment of colonial institutional frameworks or resource discoveries, which have amplified effects over time. Lock-in effects result in suboptimal equilibria due to increasing costs of reversal, leading to entrenched inefficiencies in institutions and technologies. Critical junctures are defined as pivotal moments that determine divergent developmental paths, emphasizing the role of historical exogenous shocks, such as oil discoveries or colonialism, in shaping institutional development and growth trajectories. These concepts provide insight into the mechanisms through which historical events and exogenous shocks, namely oil discoveries and colonial legacies, shape long-term institutional development trajectories. This framework illustrates how initial conditions can constrain or propel the evolution of institutions, leading to persistent inefficiencies or growth patterns that are increasingly difficult to reverse.

Gita Subrahmanyam (2006) provides further insights into the path-dependent dynamics of colonialism in the paper "Ruling Continuities: Colonial Rule, Social Forces and Path Dependence in British India and Africa." Through a comparative analysis of British India, Pakistan, and African colonies, the paper illustrates how colonial rule created entrenched institutional path dependencies, conditioning economic, social, and political development. The study found that the autocratic nature of British colonial governance negatively influenced institutional development post-independence in all three studied colony types, impeding democratic consolidation and diminishing institutional quality. Furthermore, the extractive nature of colonial governance impacted social and economic development by prioritizing resource extraction over human capital and infrastructure investment, precluding inclusive growth. These findings support the contention that colonial legacies create path dependence, a premise this study adopts to explore how these legacies interact with oil-driven institutional path dependence.

## **2.3 Economic Growth Models**

Growth is among the most researched topics in economics, as authors seek to uncover its necessary conditions to engender development and raise economic output. A substantial portion of this literature builds upon Solow's neoclassical framework, seeking to incorporate factors such as institutional quality, human capital, and natural constraints to explain variation in growth patterns. The following section examines the relevant growth literature to outline the foundations of the Solow-based model used in this paper.

Mondjeli and Tsopmo (2017) investigate the interplay of natural resources, institutional quality, and economic growth in their study "The Effects on Economic Growth of Natural Resources in Sub-Saharan Africa." Using a Panel Smooth Transition Regression (PSTR) model, the study analyzed 18 Sub-Saharan countries to assess how institutional quality mediates the relationship between natural resources and growth. The study found that the direct effect of natural resources on economic growth is negative, supporting the "Resource Curse" hypothesis. The role of institutional quality was that of a mediating force, with low institutional quality lead-

ing to negative growth effects while high-quality institutions made natural resources a driver of growth. These results were consistent with robustness tests using dynamic panel models, emphasizing the non-linear nature of the relationship between natural resources and growth. These findings not only highlight the importance of governance in growth outcomes, but also prompt further exploration into how institutional quality and resource dependence can reinforce or mitigate path dependence effects of natural endowments in developing countries.

In the study "Natural resources and economic growth in Africa: The role of institutional quality and human capital", Zallé (2019) analyzed institutional quality's influence on the growth effects of natural resources. Using an Autoregressive Distributed Lag (ARDL) model, the study analyzed data from 29 African countries between 2000 and 2015. The augmented Solow model incorporated resources, institutions, and human capital, enabling the exploration of interactions between institutions and factor inputs to explain the multidimensional impact of institutions on growth. Weak institutions were found to exacerbate the resource curse due to rent seeking and corruption, whereas high-quality institutions improved resource management to amplify growth advantages. This analysis highlights the utility of augmented Solow model in analyzing institutional interactions and its influence on growth trajectories.

Temple (1998) further analyzes the applicability of augmented Solow models in econometric research in a study titled "Robustness Tests of the Augmented Solow Model". Through a detailed evaluation of heterogeneity, outliers, and measurement error, Temple challenges the empirical validity of the results in Mankiw, Romer, and Weil's (1992) foundational work. The findings emphasize the importance of accounting for cross-country heterogeneity and measurement error in growth regressions, as these factors significantly affect estimates of convergence and growth dynamics. Temple also highlights panel data issues such as omitted variable bias, which skews growth estimations and weakens the explanatory power of the augmented Solow model in certain contexts. These findings call for refined methodologies that better account for country-specific factors and robust statistical techniques, such as heteroskedasticity-consistent standard errors, to improve the applicability of growth models.

## **2.4 Causal Identification through Instruments**

The path dependence analysis of this paper relies on establishing a causal relationship between institutional quality and economic output using oil-based instruments. Endogeneity, reverse causality, omitted variables, and measurement errors complicate the estimation of direct and indirect effects in growth models, as seen in previous literature. The instrumental variables (IV) approach addresses these challenges, offering a robust framework to isolate causal effects where randomized controlled trials are impractical. This section reviews key contributions to causal identification through instruments, focusing on their application to resource-based economics and institutional dynamics. It also details the commonly accepted practices regarding the as-

sumptions, validity, and strength conditions of instrumental variables, setting the foundation for applying these tools to path-dependent institutional dynamics.

In "Natural Resource Abundance, Institutions and Economic Growth," Brunnschweiler (2008) critically examines the resource curse through an Ordinary Least Squares (OLS) and Two-Stage Least Squares (2SLS) approach. Brunnschweiler argues that prior studies on the resource curse inadequately capture resource abundance by relying on primary exports in GDP, proposing the use of per capita resource wealth as a more accurate proxy for resource endowment. The study employs latitude as an instrumental variable, justifying its use by demonstrating that latitude influences economic development only through its impact on institutional quality. The 2SLS results suggest that institutional quality does not mediate a "curse"; rather, resources can be beneficial when paired with effective governance. Brunnschweiler demonstrates the robustness of these findings through multiple regressions and controls, highlighting the value of IV methods in revealing the complex relationships between resources, institutions, and growth.

Aleksin and Becker (2016) provide further insight into the use of IV methods, highlighting both their strengths and limitations in establishing causality in contexts where controlled trials are not feasible. The contributions of this study lie in clarifying the core IV principles, namely relevance and exogeneity. The study explains that instruments must strongly correlate with the endogenous treatment variable but must not directly affect the outcome except through the treatment variable. This method is found to effectively address endogeneity issues such as omitted variable bias, reverse causality, and measurement error. Nonetheless, the authors warn that the causal effects are identified only for the subset of the population influenced by the instrument, limiting their generalizability. The study also emphasizes the importance of addressing weak instruments, as insufficient correlation with treatment variables can lead to biased estimates. The F-statistic is highlighted as a critical diagnostic tool for assessing instrument strength, with a value of 10 being the commonly accepted threshold for a relevant instrument. This provides essential guidance for ensuring the reliability of IV results, in line with established econometric standards and best practices in economic research.

The study "Weak Instruments in Instrumental Variables Regression" by Andrews et al. (2019) delves into the issue of weak instruments and its implications for reliable causal inference in empirical research. The authors survey recent studies and demonstrate the prevalence of weak instruments in empirical research, emphasizing the need for rigorous diagnostic checks, such as the Montiel Olea & Pflueger effective F-statistic. This study provides actionable insights by advocating for weak-instrument-robust methods, and highlights the importance of careful instrument selection and testing to enhance empirical reliability.

Despite significant advances in economic research, critical gaps remain in understanding how oil wealth interacts with institutional path dependence and historical contexts, particularly colonial legacy. Building on the reviewed literature, the forthcoming analysis integrates institutional theory, path dependence, instrumental variables, and growth models to examine the

long-term effects of oil wealth on economic outcomes, focusing on how path-dependent institutional dynamics shape these effects. This thesis seeks to provide novel insights into the mechanisms that shape institutional trajectories and economic outcomes in oil-rich countries. It does so by considering two key factors: first, the role of colonial history in influencing institutional development, and second, the disaggregation of institutional quality to identify the specific dimensions impacted by oil wealth. This approach moves beyond composite measures, offering a more nuanced understanding of the institutional dynamics associated with the path-dependent impact of oil. In doing so, it contributes to the literature by focusing specifically on oil commodities while broadening the geographical scope to include developing countries beyond commonly studied regions.

### 3 Methodology

The countries examined in this paper are developing nations that have experienced significant oil reserve discoveries as pivotal moments in their growth trajectories. These countries include Saudi Arabia, Kuwait, Iran, Iraq, Algeria, Nigeria, Angola, Gabon, Equatorial Guinea, Ghana, Venezuela, Brazil, Ecuador, Mexico, Indonesia, Malaysia, Kazakhstan, and Azerbaijan. They exhibit diverse growth strategies and varying approaches to leveraging oil resources, resulting in distinct development paths over the past century. From success stories to failed states, the exogenous shocks of oil discoveries have profoundly influenced their institutional development, offering valuable insights into the dynamics of oil path dependence. The analysis spans the period from 1991 to 2020, given the limited availability of earlier data for developing nations. Data for this study are sourced from three primary datasets: the World Bank, Our World in Data, the Global Data Lab and the V-DEM democracy dataset.

The methodological framework used to analyze the path-dependent effects of oil wealth on institutional quality and economic growth align with the challenges identified in the literature. A two-stage least squares (2SLS) regression approach is employed to investigate the relationship between institutional quality and oil endowments, while establishing a causal link between institutions and economic output. To account for unobserved heterogeneity and control for country-specific and time-specific factors, panel regression is used with year and country fixed effects. Additionally, heteroskedasticity-robust standard errors are applied to ensure the reliability of the estimated coefficients, addressing potential variability in the error terms.

In the first stage, the proposed instrument—a ten-year lag of oil resource endowments—is regressed on a composite institutional measure, along with other variables from the augmented Solow model. This institutional measure consists of five key components: property rights, corruption, absence of political violence, transparent laws, and bureaucratic quality, which collectively provide a comprehensive view of institutional quality. The aggregation is done through a principal component analysis (PCA). In the second stage, the augmented Solow model is estimated with the instrumented institutional measure. The variables in the Solow model include capital, labor, fixed land, and nonrenewable resources. Robustness checks are carried out through the inclusion of potential confounding factors in a second 2SLS, namely precipitation, unemployment and temperature. This supporting model is leveraged to test the relevance of the instrumental variable, testing for alternative channels of interaction between the instrumental variable and GDP per capita.

The analysis is then extended to include interaction dummies to capture the effect of colonial history, re-estimating the second stage to test the impact of colonial type on the instrumented institutions. Finally, the composite institutional measure is disaggregated into its five individual components, each of which is analyzed separately in a set of five 2SLS regressions. In these regressions, the individual institutional components serve as the dependent variable in the first

stage, and the second stage estimates the impact of each fitted instrumental variable, offering insights into the varying magnitudes of the individual institutional factors.

The residuals from these regressions are expected to exhibit heteroskedasticity, serial correlation, and cross-sectional dependence due to the nature of panel data. Employing fixed effects in panel regression addresses unobserved time-invariant heterogeneity by allowing each cross-sectional unit, namely individual countries, to have its own intercept. Country-fixed effects enable the model to account for the historical path dependence of each nation (Acemoglu & Johnson, 2005). Year-fixed effects further ensure that global shocks or trends, such as fluctuations in commodity prices, are not driving the results (Bröderl & Ludwig, 2014). For additional robustness, heteroskedasticity-consistent standard errors (HCSE) are employed, ensuring that the estimated coefficients remain unbiased and statistically reliable, despite the complexities inherent in panel data.

## Overview of Research Methodology

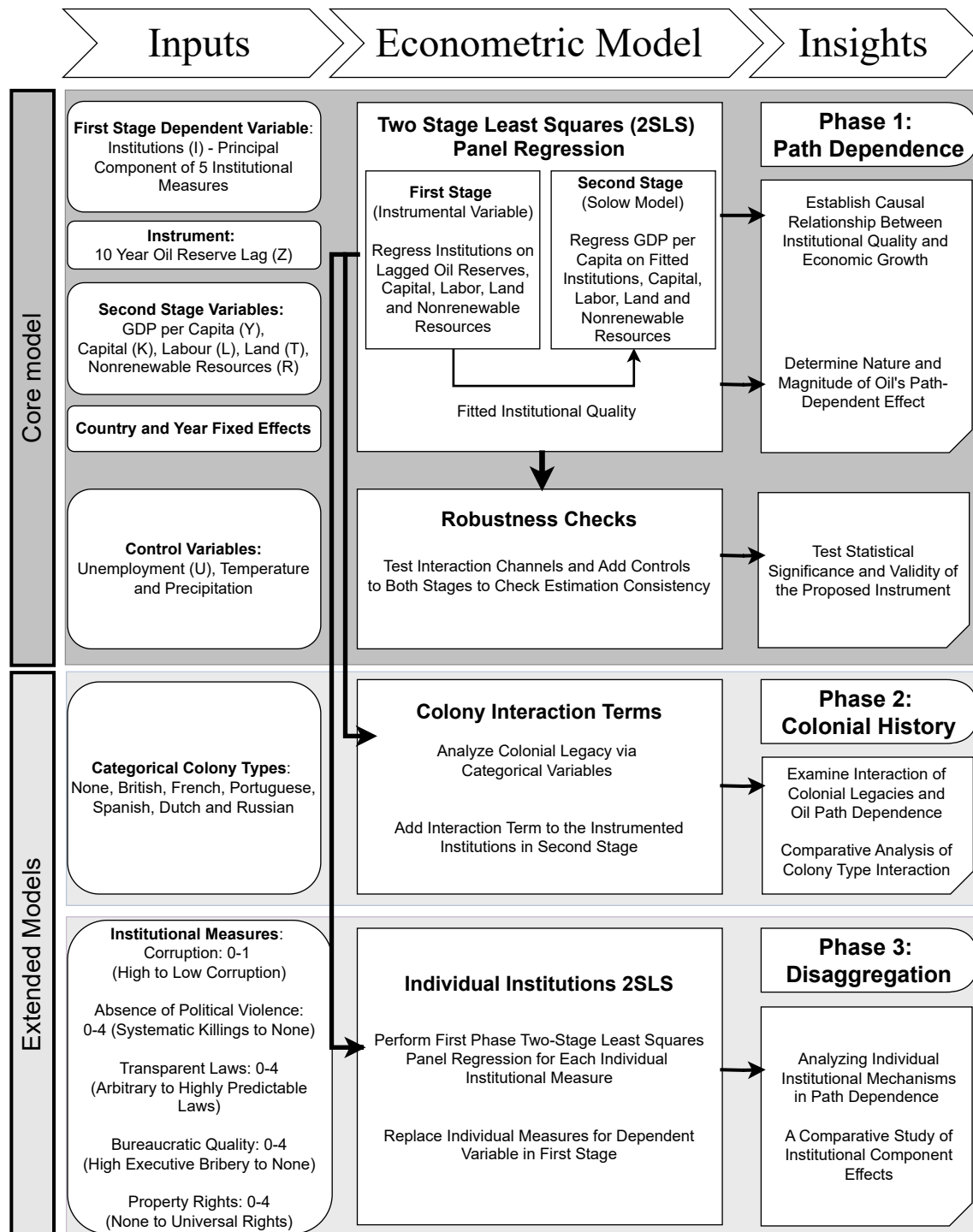


Figure 1: Methodology Flowchart

### 3.1 Data

The composite institutional measure used in this paper encompasses corruption, absence of political violence, property rights, bureaucratic quality, and transparent laws, all sourced from the V-Dem dataset (Coppedge et al., 2024). The political corruption index (code v2x\_corr in the dataset) measures the prevalence of corruption within a regime, with values ranging from 0 (less corrupt) to 1 (more corrupt). For consistency with other measures where higher values indicate more favorable outcomes, this index is inverted in the dataset. As a result, 0 and 1 represent high and low corruption, respectively.

Absence of political violence, proxied by freedom from political killings (v2clkill), answers the question "Is there freedom from political killings in a country?" This includes violence perpetrated by agents of the state, such as armed forces, police, prison officials, or any other state actors, without due legal process. It is scaled from 0 (highly systematic political killings) to 4 (fully respected, non-existent political killings). Property rights are taken as the average of property rights for men (v2clprptym) and property rights for women (v2clprptyw). Both indices are ordinal, ranging from 0 (no private property of any kind) to 5 (all or almost all individuals enjoy property rights).

Bureaucratic quality is represented as the average of two measures: executive bribery and corrupt exchanges (v2exbribe) and executive embezzlement and theft (v2exembezt). The bribery index measures how often bribes are expected in exchange for favors, with values ranging from 0 (routinely expected) to 4 (rarely or never occur). The embezzlement index reflects the frequency of public funds being stolen, misappropriated, or embezzled by executive officials, also scaled from 0 (constant embezzlement) to 4 (rarely or never occur). Finally, transparent laws with predictable enforcement (v2cltrnslw) measure how clear, coherent, and well-publicized laws are and whether their enforcement is consistent. This is an ordinal scale running from 0 (non-existent or arbitrarily enforced laws) to 4 (highly transparent and predictable enforcement).

These measures were chosen as they align with the literature's recommendations for assessing institutions in developing economies. Leftwich (1993) emphasizes that conventional metrics, such as democracy, do not fully capture the institutional quality necessary for development in these contexts. For example, nations like Saudi Arabia and the UAE, which operate under monarchical or non-democratic systems, would score poorly on conventional democracy indices. This would lower their institutional quality scores and misrepresent their ability to leverage oil wealth. By including alternative measures, namely property rights, bureaucratic quality, and the rule of law, the analysis avoids this bias and provides a more accurate representation of the institutional dimensions influencing development and growth in these developing economies.

Table 1: Institutional Variables

| Variable                                    | Description                                                                                                       |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Political Corruption (v2x_corr)             | Level of corruption (higher indicates less corruption, scale from 0 to 1)                                         |
| Absence of Political Violence (v2clkill)    | Freedom from political killings (higher indicates less violence, scale from 0 to 4)                               |
| Property Rights (v2clprptym, v2clprptyw)    | Average property rights (higher indicates secure property rights, scale from 0 to 4)                              |
| Bureaucratic Quality (v2exbribe, v2exembez) | Executive bribery and embezzlement (higher indicates better quality, scale from 0 to 4)                           |
| Transparent Laws (v2cltrnslw)               | Transparency, predictability and enforceability of laws (higher indicates better transparency, scale from 0 to 4) |
| Aggregate Institutions (I)                  | Principal Component Analysis of five measures (combined score)                                                    |

The Solow model utilizes data from the World Bank database (The World Bank, 2024). GDP per capita (code NY.GDP.PCAP.KD) is measured in constant 2015 US dollars. Capital is proxied by gross fixed capital formation in current US dollars (NE.GDI.FTOT.CD), while labor is represented by the total labor force (SL.TLF.TOTL.IN), which includes individuals aged 15 and older who contribute to the production of goods and services. Nonrenewable resources are captured by energy depletion in current US dollars (NY.ADJ.DNGY.CD), calculated as adjusted savings from the depletion of coal, crude oil, and natural gas. Fixed land is represented by arable land in hectares (AG.LND.ARBL.HA).

For the robustness tests of instrument validity, unemployment data is included, measured as the total unemployment rate (SL.UEM.TOTL.ZS) expressed as a percentage of the total labor force. This percentage is multiplied by the labor force indicator to present unemployment in absolute levels. Temperature and precipitation data are sourced from the Global Data Lab (van Geffen, 2023). Temperature is measured as the yearly average surface temperature in degrees Celsius, while precipitation data reflects the total yearly precipitation in meters.

Table 2: Second Stage Model Variables

| Variable                                | Description                             |
|-----------------------------------------|-----------------------------------------|
| Institutions                            | Aggregate (PCA) of Institutions ( $I$ ) |
| GDP per Capita (NY.GDP.PCAP.KD)         | Output per person ( $Y$ )               |
| Capital (NE.GDI.FTOT.CD)                | Physical capital ( $K$ )                |
| Labor (SL.TLF.TOTL.IN)                  | Total labor force ( $L$ )               |
| Nonrenewable Resources (NY.ADJ.DNGY.CD) | Resource depletion ( $R$ )              |
| Land (AG.LND.ARBL.HA)                   | Arable land ( $T$ )                     |
| Unemployment (SL.UEM.TOTL.ZS)           | Unemployment ( $U$ )                    |
| Precipitation                           | Yearly precipitation (PE)               |
| Temperature                             | Average surface temperature (TE)        |

For the second section of the analysis, countries are classified by their colonial history type. Interaction dummies are used in the second stage of the regression to estimate the impact of colonial history on the instrumented institutional effect. Each country is assigned a colonial category based on its historical context. The following table provides the classification of countries by their colonial type:

Table 3: Colony Classification by Country

| Colonial Type | Countries                                     |
|---------------|-----------------------------------------------|
| None          | Saudi Arabia, Iran                            |
| British       | Kuwait, Iraq, Nigeria, Ghana                  |
| French        | Algeria, Gabon                                |
| Portuguese    | Angola, Brazil                                |
| Spanish       | Equatorial Guinea, Venezuela, Ecuador, Mexico |
| Dutch         | Indonesia, Malaysia                           |
| Russian       | Kazakhstan, Azerbaijan                        |

The instrumental variable, a 10-year lag of oil reserves, is derived from the Our World in Data dataset (Energy Institute, 2024). This dataset measures proven oil reserves in tonnes, defined as quantities of oil that are reasonably certain to be recoverable under existing economic and operational conditions. Using proven reserves rather than total resources provides a more accurate reflection of potential economic activity and extraction efforts, capturing the growth potential and investment activities likely to be carried out. The dataset is compiled from various third-party sources, including World Oil, the Oil & Gas Journal, the OPEC Secretariat, and other databases.

### 3.2 Institution Aggregation

To construct the principal component of the institutional measure, a principal component analysis (PCA) is applied to the matrix of the five individual institutional components, defined as:

$$\mathbf{X} = \begin{bmatrix} I_{\text{corruption}} & I_{\text{violence}} & I_{\text{rights}} & I_{\text{bureaucracy}} & I_{\text{transparent laws}} \end{bmatrix},$$

where each  $I_{\text{component}}$  represents a specific institutional measure. PCA analyzes the covariance structure of  $\mathbf{X}$  by decomposing its covariance matrix,  $\Sigma$ , into eigenvalues and eigenvectors. Specifically:

$$\Sigma \mathbf{v}_k = \lambda_k \mathbf{v}_k,$$

where  $\Sigma$  is the covariance matrix of  $\mathbf{X}$ ,  $\lambda_k$  is the  $k$ -th eigenvalue, and

$$\mathbf{v}_k = \begin{bmatrix} v_{k,\text{corruption}} & v_{k,\text{violence}} & v_{k,\text{rights}} & v_{k,\text{bureaucracy}} & v_{k,\text{transparent laws}} \end{bmatrix}^T$$

is the corresponding eigenvector.

The first principal component is the linear combination of the institutional components associated with the largest eigenvalue,  $\lambda_1$ . The weights  $v_{1,\text{corruption}}, v_{1,\text{violence}}, \dots, v_{1,\text{transparent laws}}$  reflect the relative contribution of each institutional component to the first principal component. By summarizing the maximum variance in the institutional measures, this approach produces a robust composite measure of institutional quality. This interpretable measure allows the main model to capture overall institutional dynamics of the developing economies, providing a proxy for the broader institutional framework that influences development and growth, and may be significantly shaped by oil wealth.

### 3.3 Augmented Solow Model

The Solow model used in the first analysis is based on the specification found in Mondjeli and Tsopmo's study of economic growth and natural resources in Sub-Saharan Africa (Mondjeli & Tsopmo, 2017). The augmented Solow model integrates physical capital, labor, land, nonrenewable resources, and institutions as fundamental factors of production. In this section, the institutional measure will be instrumented using the instrumental variable in the second stage of the 2SLS analysis.

The production function is:

$$Y(t) = K(t)^\alpha L(t)^\beta T(t)^\gamma R(t)^\delta I(t)^\zeta, \quad (1)$$

where  $Y(t)$  is GDP per capita,  $K(t)$  is physical capital,  $L(t)$  is total labor force,  $T(t)$  is arable land,  $R(t)$  is nonrenewable resources, and  $I(t)$  is the institutional measure. Taking the natural logarithm, the log-linearized version is adapted in order to fit the econometric setup:

$$\ln Y = \alpha \ln K + \beta \ln L + \gamma \ln T + \delta \ln R + \zeta \ln I. \quad (2)$$

The regression model is estimated using two-stage least squares (2SLS). The second-stage regression is:

$$\ln Y = \alpha_0 + \alpha_1 \hat{I} + \alpha_2 \ln K + \alpha_3 \ln L + \alpha_4 \ln T + \alpha_5 \ln R + \text{Fixed Effects (Year, Country)} + \epsilon, \quad (3)$$

The dynamic behavior of the variables is given by:

$$\begin{aligned}
\dot{K}(t) &= sY(t) - \delta K(t), \\
\dot{L}(t) &= nL(t), \\
\dot{R}(t) &= -bR(t), \\
\dot{T}(t) &= 0.
\end{aligned} \tag{4}$$

Here, capital accumulation depends on investment and depreciation, as a fraction of output is saved and invested, driving up capital, while the wear and depletion of capital causes the stock to decrease over time. Labor growth depends on the exogenous rate ( $n$ ) of population growth, while resource depletion decreases over time at the depletion rate ( $b$ ). Here, depletion rates reduce the availability of resources over time, impacting production dependent on  $R(t)$ , which is crucial in economies with significant extractive sectors. Lastly, fixed land, proxied by arable land, is assumed to remain constant over time, reflecting each nation's natural constraints. This model takes into account the challenges faced by developing economies with significant primary activities, emphasizing the central role of natural resource extraction in shaping both institutional frameworks and output growth.

At the steady state:

$$\begin{aligned}
g_Y &= \alpha g_K + \beta g_L - \delta g_R + \zeta g_I, \\
g_Y &= \frac{\beta n - \delta b + \zeta g_I}{1 - \alpha}.
\end{aligned} \tag{5}$$

where  $g_Y$  is the growth rate of output,  $g_K$  is the growth rate of capital,  $g_L$  is the growth rate of labor,  $-\delta g_R$  accounts for resource depletion, and  $\zeta g_I$  captures the role of institutional improvements (see Figure 5 for the full derivation).

The steady states of this model capture the broader dynamics of long-term economic evolution. The capital accumulation steady state indicates that investment offsets depreciation. This implies that even if the capital stock remains constant, output per worker or GDP per capita can still grow due to improvements in other factors, such as technological progress. The steady-state capital level depends on the savings rate ( $s$ ), depreciation rate ( $\delta$ ), and productivity ( $Y$ ). Labor grows at a constant rate ( $n$ ), meaning that output per worker depends on whether capital growth can outpace labor growth.

Institutional development, represented by  $g_I$ , captures the evolution of governance frameworks. In the steady state,  $g_I$  can vary depending on the trajectory of institutional quality. If institutions stagnate (i.e.,  $g_I = 0$ ), their contribution to economic growth ceases, leaving other factors such as capital accumulation and labor growth to drive output. However, if institutional quality improves over time (positive  $g_I$ ), it contributes directly to  $g_Y$  by enhancing the productivity of both labor and capital. This evolution underscores the need to foster institutional

development as a mechanism to drive growth and development.

In the case of resource depletion, the steady state reflects a continuous decline in nonrenewable resources over time, captured by  $-\delta g_R$ . Nonrenewable resources cannot persist indefinitely unless depletion ceases entirely (i.e.,  $b = 0$ ). This points to the importance of substituting renewable resources or diversifying economic activity to ensure sustained growth.

Lastly, Fixed land evolves at a constant value, as arable land does not expand and instead serves as a constraint on long-term growth. Historical processes, such as territorial expansions in specific regions (e.g., Kuwait after the Gulf War), may alter land availability, but such cases are exceptions rather than the rule. Addressing the fixed nature of land requires sustainable land-use policies and strategies to improve land productivity.

This analysis reflects the crucial role of institutional development in driving growth in oil-rich economies. It demonstrates that, when resource extraction is the primary economic activity, the path to long-term growth hinges on strong and evolving governance frameworks. These institutions serve as mediators, playing a part equal to other factors in determining steady-state dynamics. These steady-state dynamics demonstrate the complex relationship between oil wealth and institutional quality, showcasing how path dependence could condition economic output growth.

### 3.4 Econometric Setup

The 2SLS estimation method involves the following steps:

#### 3.4.1 First Stage

The first-stage regression equation is specified as:

$$I_{it} = \pi_0 + \pi_1 Z_{i,t-10} + \pi_2 X_{it} + \lambda_i + \tau_t + \nu_{it}, \quad (6)$$

In this specification:

- $I_{it}$  : Principal component of institutional quality for country  $i$  in year  $t$ ,
- $Z_{i,t-10}$  : Ten-year lag of proven oil resources (instrumental variable),
- $X_{it}$  : Control variables for robustness checks (includes  $\ln K_{it}$ ,  $\ln L_{it}$ ,  $\ln T_{it}$ , and  $\ln R_{it}$ ),
- $\lambda_i$  : Country-specific fixed effects,
- $\tau_t$  : Year-specific fixed effects,
- $\nu_{it}$  : First-stage error term.

This regression produces the predicted values of institutional quality,  $\hat{I}_{it}$ , which are then used in the second stage.

### 3.4.2 Second Stage

The second-stage regression estimates the causal impact of institutional quality on economic output. The equation is specified as:

$$\ln Y_{it} = \alpha_0 + \alpha_1 \hat{I}_{it} + \alpha_2 \ln K_{it} + \alpha_3 \ln L_{it} + \alpha_4 \ln T_{it} + \alpha_5 \ln R_{it} + \lambda_i + \tau_t + \epsilon_{it}, \quad (7)$$

where:

$\ln Y_{it}$  : Natural logarithm of economic output,

$\hat{I}_{it}$  : Fitted instrumental variable,

$\ln K_{it}$  : Natural logarithm of capital,

$\ln L_{it}$  : Natural logarithm of labor,

$\ln T_{it}$  : Natural logarithm of land,

$\ln R_{it}$  : Natural logarithm of nonrenewable resources,

$\lambda_i$  : Country-specific fixed effects,

$\tau_t$  : Year-specific fixed effects,

$\epsilon_{it}$  : Second-stage error term.

### 3.4.3 Robustness Check

To validate the robustness of the results, an alternative regression specification is employed, integrating unemployment ( $U$ ), precipitation ( $\phi_{it}$ ), and temperature ( $\theta_{it}$ ) as additional controls. This inclusion tests whether the coefficients in the second-stage regression change, suggesting a potential alternative impact channel for GDP per capita within the instrumental variables (IV) framework. The revised equations are as follows:

First Stage: 
$$I_{it} = \gamma_0 + \gamma_1 Z_{i,t-10} + \gamma_2 X_{it} + \gamma_3 V_{it} + \lambda_i + \tau_t + \nu_{it},$$

Second Stage: 
$$\ln Y_{it} = \beta_0 + \beta_1 \hat{I}_{it} + \beta_2 \ln K_{it} + \beta_3 \ln L_{it} + \beta_4 U_{it} + \beta_5 \phi_{it} + \beta_6 \theta_{it} + \lambda_i + \tau_t + \epsilon_{it}. \quad (8)$$

where:

|               |                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------|
| $U_{it}$      | :Unemployment as a control variable,                                                             |
| $\phi_{it}$   | :Yearly precipitation as a control variable,                                                     |
| $\theta_{it}$ | :Average surface temperature as a control variable,                                              |
| $V_{it}$      | :Additional robustness control variables (includes $U_{it}$ , $\phi_{it}$ , and $\theta_{it}$ ). |

All other variables remain consistent with the baseline specification. This two-stage estimation framework identifies the causal relationship between institutional quality and economic output while addressing endogeneity concerns. Robustness checks, including alternative specifications, enhance the credibility of the findings, and ensure comprehensive validation of the results. The additional variables serve as robustness checks to control for omitted variable bias and potential confounding factors that may influence both institutional quality and economic output, thereby improving the validity of the estimation.

The inclusion of these extra variables is grounded in their potential to influence the outcome, as they may serve as alternative channels of impact. Unemployment serves as a proxy for the health of the labor market and the efficiency of resource allocation in the economy. Higher unemployment rates could weaken labor contributions to GDP and signal inefficiencies in resource allocation. As oil production and resource rents can significantly impact labor market dynamics, resource-abundant countries may experience indirect effects of lagged oil reserves on employment levels unrelated to institutional quality.

Similarly, precipitation could indirectly influence the economy in resource-rich countries, as lagged oil reserves may correlate with GDP through climate-sensitive sectors like agriculture, bypassing the institutional channel. Likewise, temperature could affect GDP indirectly by influencing climate-sensitive economic activities. In oil-exporting countries, lagged oil reserves might be linked to economic output through the effects of extreme temperatures on labor productivity, energy demand, and infrastructure durability.

By including these control variables, the analysis ensures that the influence of oil reserves on GDP operates solely through institutional quality, safeguarding the exogeneity condition of the instrument and strengthening the robustness of the results.

#### 3.4.4 Colonial Impact

To assess the impact of colonial history on the relationship between institutional quality and economic output, dummy variables for different colony types are introduced. These dummy variables allow for the inclusion of interaction terms between the fitted institutional variable ( $\hat{I}_{it}$ ) and each type of colonial legacy. The econometric specification is as follows:

First Stage:  $I_{it} = \gamma_0 + \gamma_1 Z_{i,t-10} + \gamma_2 X_{it} + \gamma_3 V_{it} + \lambda_i + \tau_t + \nu_{it},$

Second Stage:  $\ln Y_{it} = \beta_0 + \beta_1 \hat{I}_{it} + \sum_c \delta_c \left( \hat{I}_{it} \times \text{ColonialType}_c \right) + \beta_2 \ln K_{it} + \beta_3 \ln L_{it}$   
 $+ \beta_4 \ln T_{it} + \beta_5 \ln R_{it} + \beta_6 \ln U_{it} + \lambda_i + \tau_t + \epsilon_{it}.$

(9)

where:

$\text{ColonialType}_c$  :Colonial-type dummy variables,  
 $\sum_c \delta_c (\hat{I}_{it} \times \text{ColonialType}_c)$  :Interaction terms between institutions and colony-type.

By interacting  $\hat{I}_{it}$  with each colonial type, this model identifies whether and how the effect of institutional quality on economic output varies depending on a country's colonial legacy.

### 3.4.5 Disaggregated Institutions

In this final section of empirical analysis, each institutional component is evaluated individually to assess its specific impact on economic output, with an emphasis on how the magnitude of the effect varies in relation to path dependence. The corresponding regressions are specified as follows:

First Stage:  $I_{j,it} = \pi_{0,j} + \pi_{1,j} Z_{i,t-10} + \pi_{2,j} X_{it} + \lambda_i + \tau_t + \nu_{j,it},$   
 where  $j \in \{\text{corruption, violence, rights, bureaucracy, laws}\},$

Second Stage:  $\ln Y_{it} = \beta_{0,j} + \beta_{1,j} \hat{I}_{j,it} + \beta_2 \ln K_{it} + \beta_3 \ln L_{it}$   
 $+ \beta_4 \ln T_{it} + \beta_5 \ln R_{it} + \beta_6 \ln U_{it} + \lambda_i + \tau_t + \epsilon_{j,it},$

where  $\hat{I}_{j,it}$  represents the fitted values for each institutional component.

(10)

### 3.5 Instrumental Variable

In standard OLS estimations, endogenous variables are correlated with the error term, as this leads to biased and inconsistent estimates and inference. The IV provides a solution to this issue by isolating exogenous variation in the endogenous variable, allowing for causal inference (Andrews et al., 2019). For this instrument to be valid, it must satisfy two key conditions: instrument relevance and instrument exogeneity.

To satisfy instrument relevance, the IV must be strongly correlated with the endogenous explanatory variable, ensuring it can explain a significant portion of the variation in the endogenous variable. Formally, this requires:

$$\text{Cov}(Z, I) \neq 0$$

where  $Z$  represents the instrument, and  $I$  is the endogenous explanatory variable. A weak instrument can lead to weak identification and unreliable results in two-stage least squares (2SLS) (Becker & Aleksin, 2016). Therefore, the instrument's relevance will be tested using the F-statistic in the first-stage regression.

Instrument exogeneity requires that the instrument  $Z$  affects GDP per capita  $Y$  only through its impact on institutional quality  $I$  and not directly or via omitted variables. Formally, this requires:

$$\text{Cov}(Z, \varepsilon) = 0$$

where  $\varepsilon$  is the error term in the second-stage regression. This condition ensures that any relationship between  $Z$  and  $Y$  operates exclusively through  $I$ , avoiding bias from omitted variable bias or direct effects.

In this analysis, the geological origins and randomness of oil reserves, along with the use of a 10-year lag, provide a strong argument for the plausibility of exogeneity. While overidentification tests, such as Hansen's J-test, can formally assess exogeneity, these require more than one instrument. In this case, the exogeneity assumption is supported by logical arguments about the geological and historical nature of the instrument, rather than relying solely on formal statistical tests. Oil reserves, being the result of geological processes, are plausibly exogenous. A key consideration is that the discovery of oil is an exogenous event in a country's development trajectory, essentially a random event and not one driven by internal factors. While political institutions might have influenced exploration incentives, the use of a 10-year lag mitigates the risk of contemporaneous feedback into GDP and per capita output, thus reducing concerns of reverse causality.

Essentially, the impact of oil reserves on GDP per capita is mediated through its effect on institutional quality, given the time lag. This assumes that oil wealth is more likely to influence institutions, long-term developmental processes, rather than GDP, which responds on a

year-to-year basis. The entrenched, path-dependent institutional processes that emerge from oil discoveries and persist over time provide the channel through which oil wealth affects GDP per capita in this model. Oil wealth incentivizes state capacity and institutional development or, conversely, can hinder institutional progress through rent-seeking behaviors. This assumption is supported by literature suggesting that institutions mediate the relationship between oil wealth and economic growth (Brunnschweiler, 2008), particularly when an intertemporal effect reduces the possibility of direct impact. The relationship between the variables can be illustrated as follows:

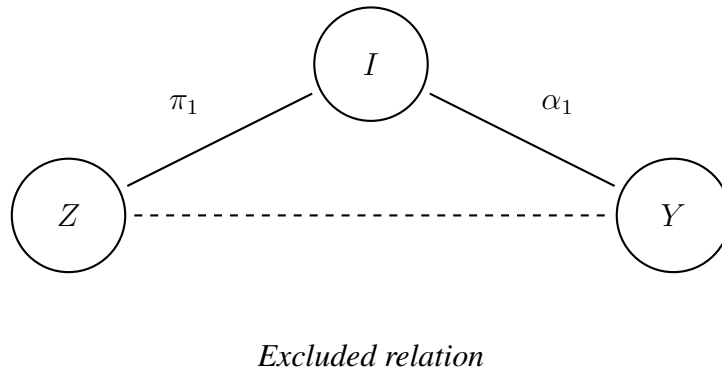


Figure 2: Instrumental Variable Exclusion Criterion

## **4 Analysis**

The analysis section presents the empirical findings, addressing the research question of whether institutions are path-dependent on oil, and whether this has causal effects on economic output.

### **4.1 Main Results**

The main regression results support the presence of institutional path dependence on oil, revealing a negative relationship. Using the 2SLS framework, the results confirm the strength and relevance of the instrumental variable, demonstrating the explanatory power of the 10-year oil lag in shaping institutions. Additionally, the findings highlight the robustness of the Solow model in the second stage of the analysis.

Table 4: 2SLS Regression

| Dependent Variables:                  | log_I                  | log_Y                  |
|---------------------------------------|------------------------|------------------------|
| Model:                                | (1)                    | (2)                    |
| <i>Variables</i>                      |                        |                        |
| 10 Year Oil Reserves Lag (log_IV)     | -0.1621***<br>(0.0379) |                        |
| Capital (log_K)                       | 0.1722***<br>(0.0378)  | 0.1740***<br>(0.0299)  |
| Labor (log_L)                         | -0.1906<br>(0.1411)    | -0.9989***<br>(0.0453) |
| Fixed Land (log_T)                    | 0.8990***<br>(0.1828)  | 0.5747***<br>(0.1578)  |
| Nonrenewable Resources (log_R)        | -0.2000***<br>(0.0396) | 0.0681<br>(0.0455)     |
| Instrumented Institutions (Fitted IV) |                        | -0.5482***<br>(0.1929) |
| <i>Fixed-effects</i>                  |                        |                        |
| Year                                  | Yes                    | Yes                    |
| Country                               | Yes                    | Yes                    |
| <i>Fit statistics</i>                 |                        |                        |
| Observations                          | 374                    | 374                    |
| R <sup>2</sup>                        | 0.89913                | 0.97523                |

*Heteroskedasticity-robust standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

The statistically significant relationship between institutions and the 10-year lag of oil reserves confirms that oil wealth systematically affects institutional quality. The negative coefficient suggests that higher oil reserves are linked to weaker institutional quality, supporting the Resource Curse hypothesis. In the second stage of the analysis, the results establish a causal link between institutional quality and GDP per capita. The fitted institutional quality variable, instrumented by the oil lag, reveals a significant negative causal impact on GDP per capita.

In the first stage regression, all variables except labor are significant at the 0.01 p-value threshold. Capital, fixed land, and nonrenewable resources notably influence institutional quality. The direction and magnitude of these effects align with economic intuition and the reviewed literature. Nonrenewable resources exhibit a negative impact of similar magnitude to the oil

variable, reinforcing the detrimental effect of natural endowments on institutions. Conversely, capital and fixed land have positive effects, consistent with economic theory, as capital accumulation is thought to correlate with higher levels of GDP per capita and overall output growth. The Solow model in the second stage yields statistically significant results, with capital, labor, and fixed land significant at the 0.01 level. The direction of effects aligns with expectations, supporting the model's robustness. Nonrenewable resources exhibit a positive coefficient in the second stage, diverging from their first-stage impact, but the effect is statistically insignificant and thus inconclusive.

The results from the 2SLS estimation confirm the strength of the instrumental variable and the explanatory power of the 10-year oil lag in determining institutional quality. The F-test value of the instrumental variable is 18.29, demonstrating the validity of the instrument by exceeding the recommended threshold for instrument strength. This robustness minimizes bias in the second stage, enabling valid causal inferences about the relationship between institutional quality and GDP per capita. The second-stage instrumented institutional variable is also statistically significant at the 1% level, further supporting this strong negative causal relationship between oil-instrumented institutions and GDP per capita.

The negative coefficient can be attributed to the influence of oil wealth on the fitted institutional variable. In the second stage, the fitted measure captures the portion of institutional quality explained by the 10-year oil lag. A negative value indicates that the oil-driven component of institutional quality is linked to lower GDP per capita. These findings suggest that oil-dependent institutional development, as reflected in the instrumental variable approach, has a detrimental impact on economic performance.

These findings suggest that oil wealth contributes to institutional weaknesses in developing economies, leading to frameworks that emphasize rent-seeking and resource control rather than economic growth. Instead of boosting GDP per capita, such institutions perpetuate corruption and inefficiency, further entrenching dependence on oil-driven governance. This illustrates the persistence of historical patterns and lock-in effects in institutional development. The findings support the contention that oil wealth exacerbates the resource curse, as entrenched institutional frameworks harm economic outcomes rather than providing a positive effect on GDP per capita. The robustness check model further bolsters the validity of these economic outcomes:

Table 5: 2SLS Regression Robustness Tests

| Dependent Variables:<br>Model:        | log_I<br>(1)           | log_Y<br>(2)           |
|---------------------------------------|------------------------|------------------------|
| <i>Variables</i>                      |                        |                        |
| 10 Year Oil Reserves Lag (log_IV)     | -0.1540***<br>(0.0444) |                        |
| Capital (log_K)                       | 0.1739***<br>(0.0338)  | 0.1561***<br>(0.0297)  |
| Labor (log_L)                         | -0.2711<br>(0.2162)    | -0.8620***<br>(0.0498) |
| Fixed Land (log_T)                    | 0.8678***<br>(0.1176)  | 0.5577***<br>(0.1534)  |
| Nonrenewable Resources (log_R)        | -0.1990***<br>(0.0291) | 0.0930*<br>(0.0461)    |
| Temperature (log_TE)                  | -0.4136<br>(0.3528)    | -0.1430<br>(0.1118)    |
| Precipitation (log_P)                 | 0.0440<br>(0.0741)     | 0.0224<br>(0.0343)     |
| Unemployment (log_U)                  | 0.0509<br>(0.0478)     | -0.0940***<br>(0.0185) |
| Instrumented Institutions (Fitted IV) |                        | -0.4923**<br>(0.1944)  |
| <i>Fixed-effects</i>                  |                        |                        |
| Year                                  | Yes                    | Yes                    |
| Country                               | Yes                    | Yes                    |
| <i>Fit statistics</i>                 |                        |                        |
| Observations                          | 374                    | 374                    |
| R <sup>2</sup>                        | 0.89980                | 0.97685                |

*Heteroskedasticity-robust standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

The inclusion of additional variables does not substantially change the previous results, as the coefficient signs remain consistent and their magnitudes show little variation. In the first stage, none of the three variables—unemployment, temperature, or precipitation—are statistically significant. In the second stage, unemployment emerges as statistically significant, while temperature and precipitation are not. The robustness of the model is supported by the fact that the inclusion of these variables neither significantly alters the coefficient of instrumented institutions nor introduces significant interactions with the instrumental variable, indicating no alternative channels through which the instrument affects GDP per capita. After including these variables, the second-stage coefficient of instrumented institutions shifts slightly from -0.5482 to -0.4923, showing minimal change. Based on these findings, unemployment is the only variable with some interaction with the instrumental variable and should thus be included in future analyses to better capture the dynamics of path dependence.

Overall, the first stage of the analysis provides strong evidence of oil path dependence in the institutions of developing nations. The robust empirical findings indicate a negative effect of oil endowments on institutional quality. The results suggest that nations in the dataset developed weaker institutions due to the presence of oil, irrespective of their economic reliance on these resources. Oil thus acts as an exogenous factor that conditions institutional development and quality, affecting GDP per capita through its corrosive influence on institutions. This aligns with economic intuition, as rent-seeking behavior spurred by natural resource wealth negatively impacts institutional quality in both informal and formal spheres, particularly in developing nations with less mature bureaucratic processes and inefficient regulatory frameworks. The detrimental impact of oil extends beyond the macroeconomic level, as outlined by the export-led Dutch disease thesis, to the microeconomic level, where it affects institutional quality and economic agents, ultimately harming GDP per capita.

It is important to note that the countries in this dataset are all developing nations, starting from a baseline of weaker institutional development. As such, oil likely exacerbates pre-existing institutional weaknesses or inhibits development due to this fragile environment. In contrast, the effect of oil might differ in countries with more robust institutional frameworks. This contention is supported by the literature (Zallé, 2019) and suggests the need for further research. A comparative study including developed countries that discovered oil after their institutions were already well-established could shed light on oil path dependence in a different context. In such cases, oil might have a less significant influence on growth trajectories, given that these countries would already possess relatively strong economic and institutional foundations. Overall, the first stage of the analysis provides results that encourage further empirical investigation into the relationship between oil, institutions, and economic outcomes.

## 4.2 Colonial Impact

The initial 2SLS results can be expanded by accounting for individual colonial histories and the unique dynamics of colonial path dependence. Introducing interaction terms for different types of colonies allows for a deeper analysis of path dependence and helps explore whether oil-related path dependence interacts with the specific colonial legacies of each country.

Table 6: 2SLS Regression Colony Type

| Dependent Variables:             | log_I                  | log_Y                  |
|----------------------------------|------------------------|------------------------|
| Model:                           | (1)                    | (2)                    |
| <i>Variables</i>                 |                        |                        |
| Proposed Instrument (log_IV_oil) | -0.1578***<br>(0.0378) |                        |
| Capital (log_K)                  | 0.1754***<br>(0.0395)  | 0.1905***<br>(0.0256)  |
| Labor (log_L)                    | -0.2462<br>(0.1645)    | -0.7395***<br>(0.0592) |
| Fixed Land (log_T)               | 0.8829***<br>(0.1779)  | 0.7732***<br>(0.1202)  |
| Nonrenewable Resources (log_R)   | -0.2048***<br>(0.0420) | 0.0446<br>(0.0305)     |
| Unemployment (log_U)             | 0.0406<br>(0.0492)     | -0.0997***<br>(0.0215) |

|                                                           |            |         |
|-----------------------------------------------------------|------------|---------|
| <i>Colony Interaction Variables</i>                       |            |         |
| Instrumented Institutions (Fitted IV) × British Colony    | -0.9746*** |         |
|                                                           | (0.1479)   |         |
| Instrumented Institutions (Fitted IV) × Dutch Colony      | -0.3567*** |         |
|                                                           | (0.1209)   |         |
| Instrumented Institutions (Fitted IV) × French Colony     | -1.260***  |         |
|                                                           | (0.1427)   |         |
| Instrumented Institutions (Fitted IV) × No Colony         | -0.9087*** |         |
|                                                           | (0.1743)   |         |
| Instrumented Institutions (Fitted IV) × Portuguese Colony | -1.234***  |         |
|                                                           | (0.1927)   |         |
| Instrumented Institutions (Fitted IV) × Russian Colony    | -0.7443*** |         |
|                                                           | (0.1910)   |         |
| Instrumented Institutions (Fitted IV) × Spanish Colony    | -0.7305*** |         |
|                                                           | (0.0965)   |         |
| <i>Fixed-effects</i>                                      |            |         |
| Year                                                      | Yes        | Yes     |
| Country                                                   | Yes        | Yes     |
| <i>Fit statistics</i>                                     |            |         |
| Observations                                              | 374        | 374     |
| R <sup>2</sup>                                            | 0.89930    | 0.98322 |

*Heteroskedasticity-robust standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

The interaction between instrumented institutions and colonial legacies in the second stage demonstrates that the relationship between oil-dependent institutions and GDP per capita varies across countries with different colonial histories. The negative and statistically significant coefficients for all interaction terms confirm that oil has a detrimental effect on institutional development and reinforces the negative causal relationship between oil-instrumented institutions and GDP per capita, regardless of the colonial legacy. However, the magnitude of this impact varies, with certain colonial legacies either amplifying or mitigating the effect.

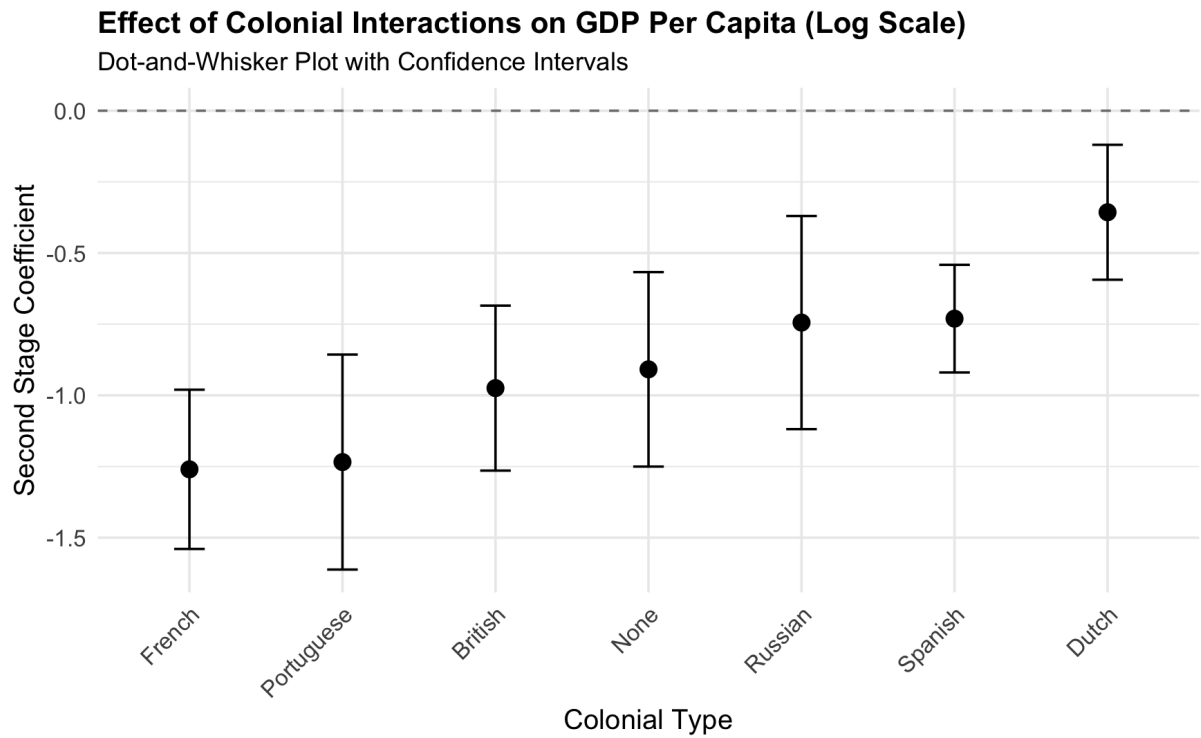


Figure 3: Colony Type Interaction

French colonies exhibit the most negative impact, with a coefficient of -1.270, followed closely by Portuguese colonies at -1.234, both significantly more negative than the original 2SLS coefficient of -0.492. The extractive institutions established during French and Portuguese colonial rule, as described by Acemoglu, Johnson, and Robinson (2001), worsen the negative influence of oil presence, intensifying path dependence dynamics. These extractive systems prioritized resource extraction and trade to benefit the colonial power, neglecting institutional development. This laid the groundwork for political instability and economic stagnation in the postcolonial period (Jerónimo, 2018); (Tadei, 2018).

Countries with no colonial history, such as Saudi Arabia and the UAE, show a moderate impact of oil path dependence, situated in the middle of the spectrum. This finding suggests that oil path dependence is not exclusive to former colonies but remains significant in shaping institutional dynamics. British colonies, by contrast, exhibit a similar moderate impact, reflecting a limited interaction between their colonial structures and oil path dependence. These results suggest that not all colonial legacies were as extractive, and in some cases, the effect of oil on institutional development may follow channels unrelated to colonial history, especially in nations that were not explicitly colonized or experienced only partial colonial influence.

Russian and Spanish colonies exhibit coefficients lower than the other colony interactions but still higher than the initial regression results. Spanish colonies, despite their extractive governance systems, appear to show a smaller interaction with oil path dependence. Lange, Mahoney, and vom Hau (2006) describe Spanish colonial governance as exceptionally extractive

and detrimental to postcolonial development, particularly when compared to British colonies. These results do not align with expectations based on the literature and the characteristics of Spanish colonial rule, which may be due to the specific countries analyzed in this study. For example, the results might reflect the limited number of British colonies in the dataset, such as Nigeria and Ghana, which were colonized for long periods. In contrast, countries like Kuwait, a protectorate, and Iraq, colonized for only 12 years, may have had insufficient exposure to fully develop the institutional path-dependent dynamics typically associated with British governance. Such limited exposure may have been insufficient to fully instill the institutional path-dependent dynamics typically associated with British governance, and thus may reflect other idiosyncratic variations in the results that are unrelated to British colonial structures.

Russian colonies, which experienced domination under the Russian Empire, Soviet influence, and later modern Russian political control, did not follow the typical extractive colonial format. Instead, they were subject to political repression and economic exploitation (Göksel & Huseynova, 2024). This indirect form of influence likely did not create the same entrenched institutional cleavages that undermined governance and development in other contexts. For instance, Azerbaijan experienced neocolonial structures prioritizing political and economic influence without direct intervention, whereas Kazakhstan suffered from forced collectivization and agricultural extraction, which stunted development. These contrasting dynamics likely explain the dampened average impact of Russian colonial legacies, producing a more moderate coefficient.

Dutch colonies stand out with the smallest negative coefficient in the instrumented institutions' impact on GDP per capita, at 0.3567, indicating a significantly mitigated effect of oil path dependence. Dutch colonial governance appears to have better managed the institutional challenges caused by oil presence in their development trajectories. The relatively higher quality of institutions, particularly in trade and commerce, may explain this relationship, providing a stronger foundation for resource management. The Dutch East India Company, for example, was highly structured and focused on trade efficiency, with an emphasis on legal institutions that supported governance (Weststeijn, 2014). Dutch colonialism, deeply tied to resource extraction, commerce, and shipping, may have established institutions that were better equipped to integrate resource wealth, such as oil, into the broader economic framework. This adaptability may have mitigated the resource curse, enabling greater institutional resilience and growth through trade. However, it is important to note that not all studies support the view that Dutch colonialism fostered efficient governance. Elise van Nederveen (2019) describes Dutch imperialism as extraordinarily extractive, with forced cultivation and coerced labor severely damaging indigenous governance and local populations. While Dutch colonial structures may have been more organized or efficient than those of other powers, their long-term impact on institutional quality and development remains contested.

It is necessary to clarify that these effects are, in many cases, based on a few countries per

colonial type. Not all colonies are included for all of the colonial type, meaning much is omitted that would better reflect the extractive nature of the colony systems. Furthermore, each colony has various idiosyncratic factors that differ even within the same colonial power, from post-colonial trajectories to endowment discoveries in pre-and post-colonial eras. As such, while these results highlight broad patterns of oil path dependence influenced by colonial structures, they should not be viewed as definitive predictors for all nations formerly under these colonial powers. This analysis is meant to shed light on the interaction of the general structure employed by each colonial power, influencing and nudging the oil path dependence dynamics.

### **4.3 Institution Disaggregation**

The purpose of disaggregating institutions is to identify the specific pathways most affected by oilpath dependence. The results for each two-stage least squares, shown in Tables 7, 9, 10, 11 and 12, indicate that the instrumental variable of lagged oil reserves is significant in the first stage for all variables except property rights. The significant coefficients are negative, in line with the core model's results. Among the variables, the magnitude of the IV is largest for the absence of political violence, indicating that oil aggravates political violence in the countries analyzed. This may be explained by the rent-seeking behavior in oil-rich countries, where state actors use political influence and force to suppress opposition and secure oil rents. This same behavior likely explains the negative effects on bureaucratic quality, corruption, and transparent laws, as oil wealth exacerbates pervasive rent-seeking and governance inefficiencies.

The insignificance of property rights suggests that oil may not directly affect their development. In many countries in the dataset, oil resources are managed mostly by the state, reducing the reliance on private property rights for resource extraction. This may render the instrument less relevant for property rights. However, this finding contrasts with studies that show oil correlates with weaker property rights (Soysa et al., 2022). It could be that in this dataset, characterized by countries with predominantly state-managed resources and autocratic tendencies, oil and property rights do not interact significantly. Alternatively, the limited observation range may fail to capture earlier periods of institutional development, suffering a reduction in degrees of freedom and statistical power.

In terms of instrument relevance, all significant institutional measures demonstrate robust first-stage results. The F-values exceed the critical threshold of 10: 12 for corruption, 15.7 for the absence of political violence, 20.4 for bureaucratic quality, and 14.3 for transparent laws. These findings strengthen the causal analysis of the disaggregated measures and validate the aggregated analysis by confirming the robustness of the majority of its individual components. Additionally, the significant interactions between oil and institutional quality highlight the relevance of these measures for analyzing developing countries and support the hypothesis of path dependence.

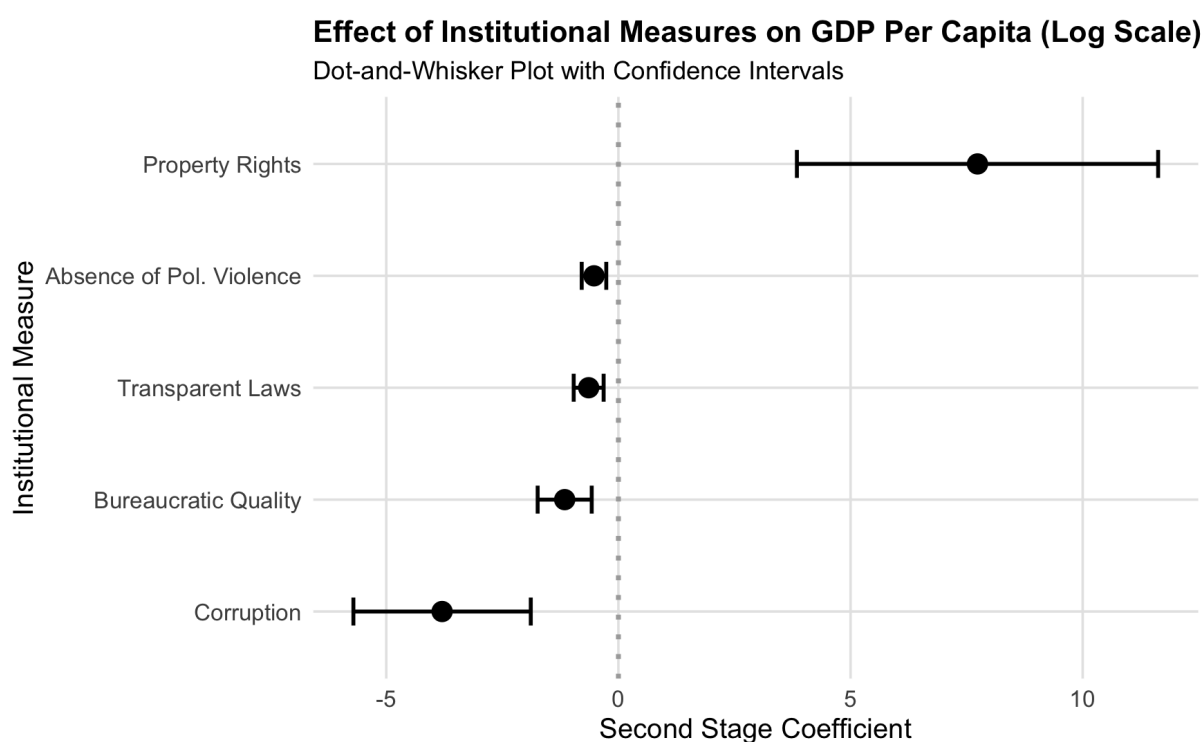


Figure 4: Institution Disaggregation

The second-stage results confirm that corruption, bureaucratic quality, transparent laws, and the absence of political violence follow the same patterns observed earlier. The negative coefficients indicate that the portion of institutional quality explained by oil (via the instrumental variable) negatively impacts GDP per capita. This suggests that oil-induced institutional degradation, particularly in governance and efficiency, stunts economic growth. Rent-seeking behavior, predatory state practices, and economic diversion driven by oil wealth undermine institutions, creating inefficiencies that have negative causal effects on GDP per capita.

Among the institutional dimensions, corruption exhibits the largest negative coefficient, suggesting it is the primary channel through which oil path dependence affects economic outcomes. Corruption fueled by oil wealth incentivizes rent-seeking, embezzlement, and favoritism, all of which erode institutional efficiency and divert resources away from productive investments.

Table 7: Corruption Disaggregated Institution

| Dependent Variables:                | log_corruption         | log_Y                  |
|-------------------------------------|------------------------|------------------------|
| Model:                              | (1)                    | (2)                    |
| <i>Variables</i>                    |                        |                        |
| 10 Year Oil Reserves Lag (log_IV)   | -0.0201***<br>(0.0058) |                        |
| Capital (log_K)                     | 0.0219***<br>(0.0054)  | 0.1536***<br>(0.0242)  |
| Labor (log_L)                       | -0.0131<br>(0.0241)    | -0.7832***<br>(0.0643) |
| Fixed Land (log_T)                  | 0.0950***<br>(0.0229)  | 0.4890***<br>(0.0997)  |
| Nonrenewable Resources (log_R)      | -0.0270***<br>(0.0063) | 0.0891***<br>(0.0342)  |
| Unemployment (log_U)                | 0.0046<br>(0.0094)     | -0.0999***<br>(0.0241) |
| Instrumented Corruption (Fitted IV) |                        | -3.798***<br>(0.9745)  |
| <i>Fixed-effects</i>                |                        |                        |
| Year                                | Yes                    | Yes                    |
| Country                             | Yes                    | Yes                    |
| <i>Fit statistics</i>               |                        |                        |
| Observations                        | 374                    | 374                    |
| R <sup>2</sup>                      | 0.91091                | 0.97684                |
| F-test                              | 69.229                 | 285.60                 |

*Heteroskedasticity-robust standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

Unlike bureaucratic inefficiencies, corruption often involves the outright theft of resources, depriving economies of funds that could otherwise be allocated to infrastructure, education, or healthcare. These diverted rents fuel elite enrichment and economic mismanagement, reinforcing cycles of institutional weakness and oil dependence. Furthermore, oil revenues, as substantial non-tax income, reduce accountability and disincentivize reform, further entrenching corruption and governance failures.

Political violence, while significant, exhibits a smaller impact compared to corruption. Its

episodic nature means it has less immediate economic repercussions, though it undermines investor confidence and stifles economic activity. Meanwhile, legal transparency and bureaucratic quality reflect longer-term governance challenges, but their indirect effects accrue over time and are less pronounced in the short term.

Property rights is the only factor with a positive second-stage coefficient, but this result should be interpreted cautiously. The first-stage F-value (0.44) and p-value above 10% indicate a weak instrument, making causal inference invalid. The observed relationship could reflect the general effect of property rights on GDP per capita, rather than the portion explained by oil wealth path dependence. This aligns with existing literature that associates secure property rights with economic growth by fostering innovation, productivity, and enterprise (Dincer, 2007).

Overall, the disaggregated results align with the findings of the literature review. They validate the selection of institutional measures for the composite analysis and provide additional insights into the relative impact of different institutional dimensions. Corruption emerges as the foremost channel of oil path dependence, exerting the most significant negative effect on GDP per capita. These results highlight the urgent need for interventions to mitigate the governance challenges posed by oil wealth, with a particular emphasis on combating corruption to promote institutional resilience and economic development.

## 5 Caveats and Study Considerations

While the results of this paper provide valuable insights into the interaction between oil, colonial legacies, institutional development, and economic growth, several limitations and considerations warrant further exploration to enhance the robustness and scope of the findings. First, a limitation of this study is the availability and range of panel data used in the analysis. The dataset spans only the last thirty years, omitting earlier periods that were critical in the development of the countries under investigation. Expanding the dataset to encompass earlier historical periods, particularly those marked by significant oil discoveries and institutional evolution, could provide a more comprehensive understanding of the long-term path dependence dynamics. Institutional change, both formal and informal, often unfolds over extended time horizons, and capturing this process requires a longer temporal perspective.

Second, the strength and validity of the instrumental variable could be improved by increasing the lag of oil reserves. Extending the lag to 20 or 30 years would address concerns of endogeneity and autocorrelation more effectively. A longer time gap would make the direct impact of the instrumental variable on the second-stage dependent variable, GDP per capita, even less plausible, thereby strengthening the robustness and validity of the instrument. The exogeneity of the instrument could also be formally tested rather than solely justified. This could be achieved by defining a second instrumental variable and conducting formal exogeneity tests, such as the Hansen J test. These tests would provide stronger evidence for the causal interpretation of the results, enhancing the study's credibility and the validity of the results.

Furthermore, including additional control variables to the robustness checks could account for alternative channels through which lagged oil reserves might impact GDP per capita, further strengthening the instrument's validity and safeguarding against omitted variable biases.

In terms of study design, the institutional measures used in the principal component analysis could be refined to capture broader aspects of path dependence and economic growth. For instance, focusing on democracy as an institutional measure while excluding authoritarian regimes could provide insights into how democratic institutions interact with oil wealth. Additionally, incorporating alternative institutional measures could highlight other dimensions of oil path dependence. Informal conventions and institutional spheres, such as social norms or informal markets, may also be significantly impacted by oil wealth. Unequal resource distribution in oil-rich countries can erode social trust and weaken informal norms of fairness and cooperation, creating a path-dependent trajectory that disrupts political stability through its social impacts. These corrosive effects on social cohesion may generate conflicts that destabilize nations, as seen during events like the Arab Spring. Investigating these dimensions could uncover less explored pathways of oil path dependence and illuminate subtle yet significant channels through which oil wealth hinders development and economic growth.

The study could also be expanded to include both developed and developing countries, in-

corporating oil-rich nations that have followed divergent growth trajectories. Including such countries would not only increase the number of observations and enhance the reliability of results but also provide valuable insights into how path dependence operates in contexts where institutional frameworks and wealth management strategies have facilitated successful economic growth. Investigating these dynamics across different income levels could shed light on how high-GDP-per-capita countries have approached resource management and institution-building to achieve sustained growth.

Finally, the empirical framework could be enhanced with advanced statistical techniques to provide deeper insights into oil path dependence dynamics. For instance, transitioning from log-level transformations to growth rates could offer a more detailed view of the growth trajectories of variables, focusing on short-term dynamic changes rather than long-term equilibrium relationships. This approach would highlight differences in oil dependence across growth trends rather than static levels of GDP per capita. Additionally, a regression discontinuity design (RDD) could be applied, specifying the date of an exogenous oil discovery as the treatment threshold. This would allow for the analysis of oil dependence and GDP per capita in distinct segments, distinguishing between amplifying and regulating mechanisms. The local treatment effect of RDD would minimize biases, allow for nonlinear relationships, and position oil discovery as a pivotal event shaping institutional development and path dependence.

## 6 Conclusion

This paper aimed to investigate the path dependence of institutions on oil in developing countries, establishing a causal relationship between institutional quality and economic output through an oil-based instrumental variable. Using a robust 2SLS framework and a relevant instrument, the findings revealed a negative path dependence of institutional quality on oil, providing a grounded answer to the study's primary inquiry.

By incorporating colonial history and disaggregating institutional measures, this study offered deeper insights into the mechanisms of oil path dependence and its detrimental impact on GDP per capita. The results highlighted corruption as the primary channel through which oil wealth erodes institutional quality and negatively affects economic output. This finding aligns with broader literature on resource-driven institutional decline. Furthermore, the enduring significance of colonial histories reinforced the persistent influence of historical legacies on institutional frameworks, enriching the existing scholarship on resource wealth and economic growth.

This analysis contributes to understanding how oil path dependence hinders institutional development and economic performance, emphasizing the critical role of institutional quality in mediating the effects of natural resource endowments. Although the methodology could be enhanced through broader data availability and including longer historical timeframes, these findings shed light on the dynamics of oil path dependence and its implications for development and policy-making. It also provides a model that can be extended to analyze various institutional components and its path-dependent dynamics, providing valuable insights into the interaction of oil wealth and factors influencing efficient governance.

As countries seek to diversify their economies and harness natural resources for economic growth, this study serves as a crucial warning that resource wealth can undermine institutional frameworks and threaten long-term development. The short-term economic gains from natural resource extraction may be outweighed by rent-seeking behaviors and governance weaknesses fostered by oil, which shape and influence institutional development paths. These insights can be extended to other natural resources, particularly energy commodities that dominate geopolitics and contemporary energy policy debates. As suggested by the results of this paper, policymakers must prioritize reducing corruption and enhancing institutional transparency to mitigate the corrosive effects of the Resource Curse. To achieve development and resilient growth, oil-rich nations must safeguard institutional frameworks to overcome oil path dependence and its hindrances, paving the way for sustainable growth.x

## A Appendix

Table 8: Description of Variables Used in the Analysis

| Variable         | Description                                                                               |
|------------------|-------------------------------------------------------------------------------------------|
| $I$              | Institutions                                                                              |
| $Y$              | Output per person                                                                         |
| $K$              | Capital                                                                                   |
| $L$              | Labor                                                                                     |
| $R$              | Resource depletion                                                                        |
| $T$              | Arable land                                                                               |
| $U$              | Unemployment                                                                              |
| $PE$ or $\phi$   | Precipitation                                                                             |
| $TE$ or $\theta$ | Temperature                                                                               |
| $Z$              | Ten-year lag of proven oil resources (instrumental variable)                              |
| $X$              | Control variables for robustness checks (includes $K$ , $L$ , $T$ , $R$ , and $\hat{I}$ ) |
| $V$              | Additional robustness control variable (includes $U$ , $\phi$ , and $\theta$ )            |
| $\lambda$        | Country-specific fixed effect                                                             |
| $\tau$           | Year-specific fixed effects                                                               |
| $\pi_1$          | First-stage coefficient of instrumental variable                                          |
| $\alpha_1$       | Second-stage coefficient of instrumented institutions                                     |
| 2SLS             | Two-stage least squares                                                                   |
| PCA              | Principal component analysis                                                              |
| IV               | Instrumental variable                                                                     |

### Steady-State Growth Rate

Starting from the production function, the growth rate of output is expressed as:

$$g_Y = \alpha g_K + \beta g_L + \gamma g_T + \delta g_R + \zeta g_I.$$

Substituting the known growth rates:

$$g_L = n, \quad g_T = 0, \quad g_R = -b,$$

the equation simplifies to:

$$g_Y = \alpha g_K + \beta n - \delta b + \zeta g_I.$$

At the steady state, the growth rate of capital equals the growth rate of output, so  $g_K = g_Y$ . Substituting this condition into the equation gives:

$$g_Y = \alpha g_Y + \beta n - \delta b + \zeta g_I.$$

Rearranging terms to isolate  $g_Y$ :

$$g_Y(1 - \alpha) = \beta n - \delta b + \zeta g_I.$$

Finally, solving for  $g_Y$  yields:

$$g_Y = \frac{\beta n - \delta b + \zeta g_I}{1 - \alpha}.$$

This is the steady-state growth rate of output, determined by labor growth, resource depletion, institutional improvement, and the contribution of capital accumulation.

Figure 5: Derivation of the Steady-State Growth Rate

Table 9: Absence of Pol. Violence Disaggregated Institution

| Dependent Variables:              | log_absence_political_violence | log_Y                  |
|-----------------------------------|--------------------------------|------------------------|
| Model:                            | (1)                            | (2)                    |
| <i>Variables</i>                  |                                |                        |
| 10 Year Oil Reserves Lag (log_IV) | -0.1452***<br>(0.0366)         |                        |
| Capital (log_K)                   | 0.1410***<br>(0.0343)          | 0.1446***<br>(0.0230)  |
| Labor (log_L)                     | -0.5921***<br>(0.1604)         | -1.045***<br>(0.0959)  |
| Fixed Land (log_T)                | 0.9333***<br>(0.1590)          | 0.6193***<br>(0.1296)  |
| Nonrenewable Resources (log_R)    | -0.1626***<br>(0.0394)         | 0.1060***<br>(0.0307)  |
| Unemployment (log_U)              | -0.0318<br>(0.0508)            | -0.1342***<br>(0.0232) |
| Instrumented Violence (Fitted IV) |                                | -0.5260***<br>(0.1350) |
| <i>Fixed-effects</i>              |                                |                        |
| Year                              | Yes                            | Yes                    |
| Country                           | Yes                            | Yes                    |
| <i>Fit statistics</i>             |                                |                        |
| Observations                      | 374                            | 374                    |
| R <sup>2</sup>                    | 0.87484                        | 0.97684                |
| F-test                            | 47.325                         | 285.60                 |

*Heteroskedasticity-robust standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

Table 10: Property Rights Disaggregated Institution

| Dependent Variables:                     | log_property_rights    | log_Y                  |
|------------------------------------------|------------------------|------------------------|
| Model:                                   | (1)                    | (2)                    |
| <i>Variables</i>                         |                        |                        |
| 10 Year Oil Reserves Lag (log_IV)        | 0.0099<br>(0.0148)     |                        |
| Capital (log_K)                          | 0.0638***<br>(0.0128)  | -0.4229***<br>(0.1351) |
| Labor (log_L)                            | 0.4141***<br>(0.0783)  | -3.936***<br>(0.8157)  |
| Fixed Land (log_T)                       | -0.0445<br>(0.0688)    | 0.4725***<br>(0.0961)  |
| Nonrenewable Resources (log_R)           | -0.1062***<br>(0.0171) | 1.013***<br>(0.2099)   |
| Unemployment (log_U)                     | 0.0568**<br>(0.0235)   | -0.5570***<br>(0.1129) |
| Instrumented Property Rights (Fitted IV) |                        | 7.734***<br>(1.985)    |
| <i>Fixed-effects</i>                     |                        |                        |
| Year                                     | Yes                    | Yes                    |
| Country                                  | Yes                    | Yes                    |
| <i>Fit statistics</i>                    |                        |                        |
| Observations                             | 374                    | 374                    |
| R <sup>2</sup>                           | 0.92137                | 0.97684                |
| F-test                                   | 79.337                 | 285.60                 |

*Heteroskedasticity-robust standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

Table 11: Bureaucratic Quality Disaggregated Institution

| Dependent Variables:<br>Model:                   | log_bureaucratic_quality<br>(1) | log_Y<br>(2)           |
|--------------------------------------------------|---------------------------------|------------------------|
| <i>Variables</i>                                 |                                 |                        |
| 10 Year Oil Reserves Lag (log_IV)                | -0.0560***<br>(0.0124)          |                        |
| Capital (log_K)                                  | 0.0901***<br>(0.0108)           | 0.1963***<br>(0.0283)  |
| Labor (log_L)                                    | 0.4015***<br>(0.0483)           | -0.7290***<br>(0.0902) |
| Fixed Land (log_T)                               | -0.0079<br>(0.0431)             | 0.6582***<br>(0.1539)  |
| Nonrenewable Resources (log_R)                   | -0.0271**<br>(0.0133)           | 0.0756***<br>(0.0373)  |
| Unemployment (log_U)                             | 0.0314<br>(0.0164)              | -0.1205***<br>(0.0292) |
| Instrumented Bureaucratic Quality<br>(Fitted IV) |                                 | 2.397***<br>(0.6292)   |
| <i>Fixed-effects</i>                             |                                 |                        |
| Year                                             | Yes                             | Yes                    |
| Country                                          | Yes                             | Yes                    |
| <i>Fit statistics</i>                            |                                 |                        |
| Observations                                     | 374                             | 374                    |
| R <sup>2</sup>                                   | 0.90234                         | 0.97684                |
| F-test                                           | 66.902                          | 285.60                 |

*Heteroskedasticity-robust standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

Table 12: Transparent Laws Disaggregated Institution

| Dependent Variables:                         | log_transparent_laws   | log_Y                  |
|----------------------------------------------|------------------------|------------------------|
| Model:                                       | (1)                    | (2)                    |
| <i>Variables</i>                             |                        |                        |
| 10 Year Oil Reserves Lag (log_IV)            | -0.0314***<br>(0.0083) |                        |
| Capital (log_K)                              | 0.0984***<br>(0.0072)  | 0.1365***<br>(0.0215)  |
| Labor (log_L)                                | 0.2561***<br>(0.0354)  | -0.6415***<br>(0.0732) |
| Fixed Land (log_T)                           | -0.0918**<br>(0.0319)  | 0.4560***<br>(0.1378)  |
| Nonrenewable Resources (log_R)               | -0.0489***<br>(0.0102) | 0.0914***<br>(0.0347)  |
| Unemployment (log_U)                         | -0.0412***<br>(0.0112) | -0.1129***<br>(0.0264) |
| Instrumented Transparent Laws<br>(Fitted IV) |                        | 4.536***<br>(1.047)    |
| <i>Fixed-effects</i>                         |                        |                        |
| Year                                         | Yes                    | Yes                    |
| Country                                      | Yes                    | Yes                    |
| <i>Fit statistics</i>                        |                        |                        |
| Observations                                 | 374                    | 374                    |
| R <sup>2</sup>                               | 0.88723                | 0.97684                |
| F-test                                       | 53.684                 | 285.60                 |

*Heteroskedasticity-robust standard-errors in parentheses*

*Signif. Codes: \*\*\*: 0.01, \*\*: 0.05, \*: 0.1*

## B Bibliography

### References

- Acemoglu, D., & Johnson, S. (2005). Unbundling institutions. *Journal of Political Economy*, 113(5), 949–995. <https://doi.org/10.1086/432166>
- Acemoglu, D., Johnson, S., & Robinson, J. A. (2001). The colonial origins of comparative development: An empirical investigation. *The American Economic Review*, 91(5), 1369–1401. Retrieved January 12, 2025, from <http://www.jstor.org/stable/2677930>
- Andrews, I., Stock, J. H., & Sun, L. (2019). Weak instruments in instrumental variables regression: Theory and practice. *Annual Review of Economics*. <https://api.semanticscholar.org/CorpusID:202126769>
- Becker, S. O., & Aleksin, G. (2016). Using instrumental variables to establish causality. *IZA World of Labor*. <https://doi.org/10.15185/izawol.250>
- Brüderl, J., & Ludwig, V. (2014, January). Fixed-effects panel regression.
- Brunnschweiler, C. N. (2008). Cursing the blessings? natural resource abundance, institutions, and economic growth. *World Development*, 36(3), 399–419. <https://doi.org/https://doi.org/10.1016/j.worlddev.2007.03.004>
- Coppedge, M., Gerring, J., Knutsen, C. H., Lindberg, S. I., Teorell, J., Altman, D., Angiolillo, F., Bernhard, M., Borella, C., Cornell, A., Fish, M. S., Fox, L., Gastaldi, L., Gjerlow, H., Glynn, A., Good God, A., Grahn, S., Hicken, A., Kinzelbach, K., . . . Ziblatt, D. (2024). V-dem [country-year/country-date] dataset v14. <https://doi.org/10.23696/mcwt-fr58>
- David, P. (2007). Path dependence: A foundational concept for historical social science. *Cliometrica*, 1(2), 91–114. <https://doi.org/10.1007/s11698-006-0005-x>
- Dincer, O. (2007). The effects of property rights on economic performance. *Applied Economics*, 39(7), 825–837. <https://doi.org/10.1080/00036840500461964>
- Ding, N., & Field, B. C. (2005). Natural resource abundance and economic growth. *Land Economics*, 81(4), 496–502. <https://doi.org/10.3368/le.81.4.496>
- Energy Institute. (2024). Statistical review of world energy (2024) – with major processing by our world in data. “oil proved reserves” [dataset] [Energy Institute, “Statistical Review of World Energy” [original data]].
- Ernst, K. M., & Preston, B. L. (2017). Adaptation opportunities and constraints in coupled systems: Evidence from the u.s. energy-water nexus. *Environmental Science Policy*, 70, 38–45. <https://doi.org/10.1016/j.envsci.2017.01.001>
- Frericks, P., Höppner, J., & Och, R. (2018). The difficulty of measuring institutions: A methodological approach to the comparative analysis of institutions. *Social Indicators Research*, 137(3), 847–865. <https://doi.org/10.1007/s11205-017-1638-9>

- Göksel, O., & Huseynova, N. (2024). The other colonial empire: Reconsidering soviet rule in the caucasus and central asia through a post-colonial lens. *Florya Chronicles of Political Economy*, 10, 211–247.
- Jerónimo, M. B. (2018, March). Portuguese colonialism in africa. <https://doi.org/10.1093/acrefore/9780190277734.013.183>
- Lange, M., Mahoney, J., & vom Hau, M. (2006). Colonialism and development: A comparative analysis of spanish and british colonies. *American Journal of Sociology*, 111(5), 1412–1462. <https://doi.org/10.1086/499510>
- Leftwich, A. (1993). Governance, democracy and development in the third world. *Third World Quarterly*, 14(3), 605–624. Retrieved January 4, 2025, from <http://www.jstor.org/stable/3992489>
- Mankiw, N. G., Romer, D., & Weil, D. N. (1992). A contribution to the empirics of economic growth. *Quarterly Journal of Economics*, 107, 407–437. [https://eml.berkeley.edu/~dromer/papers/MRW\\_QJE1992.pdf](https://eml.berkeley.edu/~dromer/papers/MRW_QJE1992.pdf)
- Mondjeli, I. M., & Tsopmo, P. C. (2017). The effects on economic growth of natural resources in sub-saharan africa: Does the quality of institutions matter? *Economics Bulletin*, 37(1), 248–263. <https://ideas.repec.org/a/eb/ecbull/eb-16-00550.html>
- North, D. C. (2005). *Understanding the process of economic change*. Princeton University Press. <http://www.jstor.org/stable/j.ctt7zvbxt>
- Ross, M. L. (2001). Does oil hinder democracy? *World Politics*, 53(3), 325–361. <https://doi.org/10.1353/wp.2001.0011>
- Soysa, I., Krieger, T., & Meierrieks, D. (2022). Oil and property rights. *Resources Policy*, 79. <https://doi.org/10.1016/j.resourpol.2022.103069>
- Subrahmanyam, G. (2006). Ruling continuities: Colonial rule, social forces and path dependence in british india and africa. *Commonwealth & Comparative Politics*, 44(1), 84–117. <https://doi.org/10.1080/14662040600624478>
- Tadei, F. (2018). The long-term effects of extractive institutions : Evidence from trade policies in colonial french africa. *Economic History of Developing Regions*, 33(3), 183–208. <https://doi.org/10.1080/20780389.2018.1527685>
- Temple, J. R. W. (1998). Robustness tests of the augmented solow model. *Journal of Applied Econometrics*, 13(4), 361–375. <http://www.jstor.org/stable/223186>
- The World Bank. (2024). World development indicators [Terms of use available at <https://doi.org/10.57966/6rwy-0b07>]. <https://doi.org/10.57966/6rwy-0b07>
- van Geffen, A. (2023). *Gdldata: Global data lab r api* [<https://docs.globaldatalab.org/gdldata/>, <https://github.com/GlobalDataLab/R-data-api>].
- van Nederveen Meerkerk, E. (2019). An exceptional empire? dutch colonialism in comparative perspective. In *Women, work and colonialism in the netherlands and java: Comparisons, contrasts, and connections, 1830–1940* (pp. 47–83). Springer International Publishing.

- Weststeijn, A. (2014). The voc as a company-state: Debating seventeenth-century dutch colonial expansion. *Itinerario*, 38(1), 13–34. <https://doi.org/10.1017/S0165115314000035>
- Zallé, O. (2019). Natural resources and economic growth in africa: The role of institutional quality and human capital. *Resources Policy*, 62, 616–624. <https://doi.org/10.1016/j.resourpol.2018.11.009>

## **C Statement of Original Thesis**

### **Official Statement of Original BACHELOR Thesis**

By signing this statement, I hereby acknowledge the submitted thesis (hereafter mentioned as “product”), titled: ”Path Dependence And Growth: The Role of Oil in Developing Economies” to be produced independently by me, without external help. Wherever I paraphrase or cite literally, a reference to the original source (journal, book, report, internet, etc.) is given.

By signing this statement, I explicitly declare that I am aware of the fraud sanctions as stated in the Education and Examination Regulations (EERs) of the SBE.

Place: Arenales 1422, Ciudad Autonoma Buenos Aires, Argentina

Date: January 20th, 2025

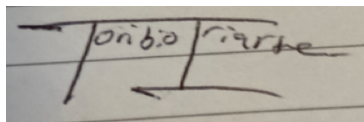
First and Last Name: Toribio Iriarte Fernandez

Study Programme: Economics and Business Economics, Economics Specialization

EBT Code: EBT0001

ID number: i6317519

**Signature:**

A photograph of a handwritten signature in black ink on a light-colored background. The signature is written in a cursive style and appears to read "Toribio Iriarte".